

CONFORMAL UNCERTAINTY FOR NO-REFERENCE MOS PREDICTION UNDER DISTRIBUTION SHIFT: A LEAKAGE-CONTROLLED STUDY

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ABSTRACT

No-reference mean opinion score (MOS) predictors return a point estimate with no signal of when to trust it, a risk for automated pipelines. We test whether distance from a reference feature distribution, measured for frozen DNSMOS P.835, NISQA, and UTMOS, can flag unreliable predictions and set per-utterance conformal interval widths. A leakage-controlled design separates feature-reference fitting, content-disjoint calibration, and evaluation across all seven NISQA Corpus subsets. Euclidean, Mahalanobis, and nearest-neighbour distances detect corpus shift (mean domain AUROC 0.615–0.690) but do not consistently flag high-error utterances within a domain (mean AUROC 0.433–0.499, near chance). Fixed split conformal intervals reach 90.1–91.8% coverage at the 90% nominal level. Mahalanobis-stratified intervals help some predictor-domain pairs and hurt others. A shift-detection score is therefore not, by itself, a reliable difficulty score for conformal interval adaptation, a finding that reproduces on an independent multilingual speech-enhancement corpus. Practitioners should treat feature-space alarms as domain-mismatch flags, not per-utterance error flags.

Index Terms— mean opinion score, conformal prediction, out-of-distribution detection, speech quality assessment, distribution shift, uncertainty quantification

1. INTRODUCTION

No-reference mean opinion score (MOS) predictors are increasingly used to screen and compare speech systems at scale. DNSMOS P.835 [1], NISQA [2], and UTMOS [3] reduce reliance on repeated large-scale listening tests, while DNSMOS also serves as an official metric in the DNS Challenge [4, 5]. Their use has outpaced scrutiny of their failure modes.

Existing MOS predictors return point estimates without an explicit reliability measure. A practitioner therefore cannot tell from the score alone whether an utterance should be trusted, reviewed, or sent to a listening test. Feature-space distance is an attractive post-hoc signal because it requires neither predictor retraining nor labelled shifted data. Its interpretation is not automatic, however. A score that recognizes a change in language, channel, or degradation process may still fail to identify which individual MOS predictions are inaccurate.

We test this hypothesis through a conformal uncertainty

wrapper for frozen MOS predictors. Split conformal prediction [6] provides finite-sample marginal coverage when calibration and test examples are exchangeable and calibration labels are independent of model fitting. The wrapper first constructs a global interval from calibration residuals. It then applies an established Mondrian-style construction [7] in which Mahalanobis distance [8] selects a distance-specific interval radius. The central question is whether a feature-space domain-shift signal transfers to utterance-level error ranking strongly enough to support this adaptation.

This work makes three contributions. First, we construct a leakage-controlled evaluation with disjoint feature-reference, conformal-calibration, and in-domain evaluation roles. Second, we compare Euclidean, Mahalanobis, and nearest-neighbour scores across the distinct tasks of domain detection and utterance-level error detection for three MOS predictors. Third, we test the practical objective directly by comparing fixed and distance-stratified conformal intervals under five metadata-defined shifts. Mean domain-detection AUROC is 0.615–0.690, while mean AUROC for detecting the highest-error quartile is only 0.433–0.499. Distance-stratified interval adaptation is predictor and domain dependent. The contribution is therefore both methodological and empirical, a controlled test showing why an OOD score must be validated as a signal of prediction error, not merely of domain membership, before it is used to adapt prediction uncertainty.

2. RELATED WORK

Early no-reference speech quality estimators modelled distortion type and signal statistics [9]. NISQA [2] uses self-attention trained on multi-condition MOS data, DNSMOS P.835 [1] predicts speech, background, and overall quality by extending the earlier single-score DNSMOS metric [10], and UTMOS [3] builds on self-supervised representations. Their development data cover languages, degradations, and channels, which motivates explicit cross-domain evaluation.

OOD detection includes confidence-based [11], density-based, and distance-based methods. Lee et al. [8] use class-conditional Mahalanobis distance for image classifiers. Deep nearest-neighbour distance has also been evaluated for audio OOD detection [12], following the general deep k -NN formulation [13], while outlier-exposure training offers a retraining-based alternative for audio OOD detection [14]. We study

a regression analogue based on one Gaussian and compare it with Euclidean and nearest-neighbour scores, without retraining the underlying predictor. Unlike conventional OOD evaluation, we separately test whether domain discrimination transfers to prediction-error ranking.

Conformal prediction converts residual scores into intervals with finite-sample marginal coverage under exchangeability [15, 6, 16]. Mondrian conformal regressors bin a difficulty estimate and calibrate a radius within each bin [7]. Weighted conformal prediction restores a coverage guarantee under a known covariate shift [17], but requires an importance-weight estimate that we do not attempt here. Recent MOS work predicts aleatoric and epistemic uncertainty for selective prediction and domain detection [18], or learns a predictive MOS distribution and filters uncertain samples [19]. We do not present distance binning, audio OOD detection, or MOS selective prediction as new methods. The novelty lies in testing, without retraining or shifted labels, whether internal feature distance transfers across domain detection, utterance-level error ranking, and conformal interval adaptation in frozen speech-quality predictors.

3. METHOD

For a speech waveform $\mathbf{x}$, a frozen no-reference predictor f returns the scalar MOS estimate $\hat{q} = f(\mathbf{x})$. We extract a fixed-dimensional representation $\mathbf{z} = \phi(\mathbf{x}) \in \mathbb{R}^d$ from the predictor and mean-pool frame- or segment-level activations when required. Each predictor has its own feature space and therefore its own reference distribution.

3.1. Feature-Space Scores

For each predictor, T-SIM supplies reference features $\{\mathbf{z}_i\}_{i=1}^N$, where $N = 10,000$. The maximum-likelihood empirical mean and covariance used by the implementation are

$$\hat{\boldsymbol{\mu}} = \frac{1}{N} \sum_{i=1}^N \mathbf{z}_i, \quad (1)$$

$$\hat{\boldsymbol{\Sigma}} = \frac{1}{N} \sum_{i=1}^N (\mathbf{z}_i - \hat{\boldsymbol{\mu}})(\mathbf{z}_i - \hat{\boldsymbol{\mu}})^\top. \quad (2)$$

Here $N > d$ for all predictors, with $d = 64$ for DNSMOS and NISQA ($N/d = 156$) and $d = 1024$ for UTMOS ($N/d = 9.8$). We use the empirical covariance throughout and do not apply Ledoit-Wolf shrinkage [20].¹ The Mahalanobis score is

$$d_M(\mathbf{x}) = \left[(\mathbf{z} - \hat{\boldsymbol{\mu}})^\top \hat{\boldsymbol{\Sigma}}^{-1} (\mathbf{z} - \hat{\boldsymbol{\mu}}) \right]^{1/2}. \quad (3)$$

¹For UTMOS, $N/d = 9.8$ is a ratio for inverting a 1024×1024 covariance matrix. Refitting this reference with Ledoit-Wolf shrinkage changes domain-detection AUROC by at most 0.09 and error-detection AUROC by at most 0.12, without changing which model has the weakest Mahalanobis domain signal or the near-chance error-detection conclusion of Section 4.3.

We also evaluate Euclidean distance $\|\mathbf{z} - \hat{\boldsymbol{\mu}}\|_2$ and the distance to the nearest T-SIM representation [13]. A larger score is prespecified to mean farther from the reference distribution. These scores quantify representation shift. They are not assumed to estimate MOS error.

3.2. Split Conformal Interval

V-SIM is partitioned by content identifier into disjoint calibration and in-domain evaluation halves of 1,250 clips each. Let $\mathcal{D}_{\text{cal}} = \{(\mathbf{x}_j, q_j^*)\}_{j=1}^m$ denote the calibration half, where $m = 1,250$ and q_j^* is subjective MOS. The absolute-residual nonconformity score is

$$s_j = |q_j^* - \hat{q}_j|, \quad \hat{q}_j = f(\mathbf{x}_j). \quad (4)$$

Let $s_{(1)} \leq \dots \leq s_{(m)}$ be the ordered calibration scores and let $\alpha \in (0, 1)$ denote the target miscoverage rate, so nominal coverage is $1 - \alpha$ (we use $\alpha = 0.10$). For $\alpha \geq 1/(m+1)$, define

$$k_\alpha = \lceil (m+1)(1-\alpha) \rceil, \quad \hat{Q}_{1-\alpha} = s_{(k_\alpha)}. \quad (5)$$

The fixed-width interval is

$$C_\alpha(\mathbf{x}) = [\hat{q} - \hat{Q}_{1-\alpha}, \hat{q} + \hat{Q}_{1-\alpha}]. \quad (6)$$

We do not clip the endpoints to $[1, 5]$, since clipping can alter coverage at the scale boundaries. If the calibration examples and a new example are exchangeable and calibration labels were not used to fit or select f , then split conformal prediction gives the marginal guarantee

$$\Pr\{q^* \in C_\alpha(\mathbf{x})\} \geq 1 - \alpha. \quad (7)$$

The content-disjoint split prevents direct content leakage but does not itself prove exchangeability, which can in principle be tested online [21]. We do not conduct that test here. The NISQA model was developed using NISQA Corpus partitions [2], so its intervals are treated as empirical rather than as a fresh-data guarantee.

3.3. Distance-Stratified Interval

Following Mondrian conformal regression with binned difficulty estimates [7], we test whether Mahalanobis distance supports instance-dependent widths. We select Mahalanobis distance for this test because Section 4.3 shows it is comparable to nearest-neighbour distance on domain-detection AUROC while requiring only a fixed (d, d) covariance matrix rather than a reference bank that grows with the calibration set. The calibration examples are divided into K empirical quantile bins of $d_M(\mathbf{x})$. Let I_k be the calibration indices assigned to bin k , let $m_k = |I_k|$, and let $s_{(r)}^{(k)}$ be the r th ordered residual within that bin. For $\alpha \geq 1/(m_k+1)$,

$$\ell_{\alpha,k} = \lceil (m_k+1)(1-\alpha) \rceil, \quad \hat{Q}_{1-\alpha}^{(k)} = s_{(\ell_{\alpha,k})}^{(k)}. \quad (8)$$

Table 1. NISQA Corpus subsets used in the study [2]. V-SIM is deterministically divided into disjoint calibration and in-domain evaluation halves. N is the number of clips with subjective MOS labels.

Abbrev.	Subset	Role	N
T-SIM	TRAIN_SIM	Feature reference	10 000
T-LIV	TRAIN_LIVE	Shifted evaluation	1 020
V-SIM-Cal	VAL_SIM	Calibration	1 250
V-SIM-ID	VAL_SIM	ID evaluation	1 250
V-LIV	VAL_LIVE	Shifted evaluation	200
T-FOR	TEST_FOR	Shifted evaluation	240
T-TLK	TEST_LIVETALK	Shifted evaluation	232
T-P501	TEST_P501	Shifted evaluation	240

If $b(\mathbf{x}) \in \{1, \dots, K\}$ is the bin assigned using the calibration-derived boundaries, then

$$C_{\alpha}^{\text{adapt}}(\mathbf{x}) = [\hat{q} - \hat{Q}_{1-\alpha}^{(b(\mathbf{x}))}, \hat{q} + \hat{Q}_{1-\alpha}^{(b(\mathbf{x}))}]. \quad (9)$$

Scores outside the calibration range enter the first or last bin. We use $K = 5$ and hold the boundaries and radii fixed at evaluation. Because the boundaries are estimated from the calibration covariates and shifted test examples need not be exchangeable with calibration examples, we treat Eq. (9) as a diagnostic post-hoc construction. We do not claim conditional or arbitrary-shift coverage for it.

4. EXPERIMENTS

4.1. Setup

We use all seven subsets of the NISQA Corpus [2], summarized in Table 1. T-SIM fits the feature reference statistics and supplies the k -NN reference bank. A deterministic content-disjoint split assigns 1,250 V-SIM clips to conformal calibration and 1,250 to in-domain evaluation. T-LIV, V-LIV, T-FOR, T-TLK, and T-P501 are evaluated as metadata-defined shifts in recording condition, language, or channel. No label from a shifted subset is used for calibration.

We evaluate DNSMOS P.835 [1] using its overall-quality output, NISQA v2 [2], and UTMOS [3]. All outputs are represented on the 1–5 MOS scale. DNSMOS predictions include the non-personalized P.835 polynomial mapping.² UTMOS scores come from a torchaudio reimplementation of the released checkpoint rather than the pinned fairseq scorer, and we have not verified numerical equivalence between the two.

For point predictions, we report SRCC and RMSE. Domain detection compares each shifted subset with held-out V-SIM using AUROC and FPR at 95% TPR. V-SIM has no domain-detection bar because it defines the in-domain negative class. Error detection is a separate within-domain task. In each

²The stored clip-level correction approximates segment-wise mapping for multi-segment clips, with mean absolute discrepancy 0.0038 MOS in a stratified verification sample.

Table 2. Point-prediction performance. Higher SRCC and lower RMSE are better. Within each dataset and metric, **bold** and underlined entries are best and second best according to unrounded values. [†]Entries that round to the same displayed value are ranked by an unrounded difference below 0.001.

Model	Metric	Held-out ID	Shifted evaluation				
		V-SIM	T-LIV	V-LIV	T-FOR	T-TLK	T-P501
DNSMOS	SRCC	0.703	<u>0.674</u>	<u>0.674</u>	0.541	0.593	0.695
	RMSE	<u>0.854</u>	<u>0.549</u>	0.525 [†]	0.845	<u>0.765</u>	0.913
NISQA	SRCC	<u>0.805</u>	0.649	0.638	<u>0.725</u>	<u>0.706</u>	<u>0.831</u>
	RMSE	0.860	0.528	<u>0.525</u> [†]	<u>0.734</u>	0.747	<u>0.653</u>
UTMOS	SRCC	0.830	0.702	0.749	0.769	0.773	0.880
	RMSE	0.662	0.829	0.852	0.604	1.022	0.636

evaluation subset, positives are the highest absolute-error quartile and the score direction is fixed in advance so that larger distance predicts larger error. Figure 1 reports group-bootstrap 95% confidence intervals from 2,000 resamples. Interval comparisons independently resample calibration and evaluation groups, then refit quantiles, bin boundaries, and bin-specific radii in every bootstrap replicate.

4.2. Point Prediction

Table 2 shows that a metadata-defined shift does not imply uniform degradation. NISQA was trained on partitions of this corpus, so its numbers are an in-sample reference rather than a generalization test comparable to DNSMOS and UTMOS. UTMOS has the highest SRCC on every subset but reaches RMSE 1.022 on T-TLK. DNSMOS attains nearly the same V-LIV RMSE as NISQA, yet has substantially higher error on T-FOR and T-P501. Rank correlation and scale error therefore provide different views of transfer. Averaging the three point estimates raises SRCC over every single model on both corpora. On URGENT2025, the ensemble mean reaches SRCC 0.674 versus at most 0.597 for any single model.

4.3. Domain Detection Versus Error Detection

Table 3 evaluates domain discrimination. Averaged over 15 predictor–shift pairs, AUROC is 0.615 for Euclidean distance, 0.678 for Mahalanobis distance, and 0.690 for nearest-neighbour distance. Nearest-neighbour distance attains the highest per-predictor AUROC in all three columns, though the margin over Mahalanobis is small for NISQA and negligible for UTMOS, while Mahalanobis obtains the lowest mean FPR@95. Neither score dominates on both metrics, so we treat Mahalanobis and nearest-neighbour distance as comparable alternatives for domain detection.

Figure 1 shows that domain discrimination and error detection are not interchangeable. Across the 18 predictor–subset pairs, mean error-detection AUROC is 0.478 for Mahalanobis distance. The corresponding means are 0.433 for Euclidean distance, 0.499 for nearest-neighbour distance, and 0.421 for prediction extremity $|\hat{q} - 3|$. On T-LIV, Mahalanobis AUROC

Table 3. Domain detection averaged across five shifted subsets against held-out V-SIM. Higher AUROC and lower FPR@95 are better. **Bold** and underlined entries are best and second best in each column according to unrounded values. [†]Entries that round to the same displayed value are ranked by an unrounded difference below 0.001.

OOD score	AUROC $\uparrow$				FPR@95 $\downarrow$
	DNS	NIQ	UTM	Avg.	Avg.
L2 Euclidean	0.585	0.693	<u>0.569</u> [†]	0.615	0.854
Mahalanobis	<u>0.668</u>	<u>0.803</u>	0.561	<u>0.678</u>	0.732
k -NN ($k = 1$)	0.697	0.805	0.569 [†]	0.690	<u>0.763</u>

Table 4. Coverage (Cov., %) and mean width (W) at 90% nominal coverage. Fixed (F) and five-bin Mahalanobis-adaptive (A) intervals use the same V-SIM calibration split. Interpret coverage and width jointly.

Dataset	DNSMOS		NISQA		UTMOS	
	F	A	F	A	F	A
	Cov.	W	Cov.	W	Cov.	W
V-SIM	90.2	2.74	90.1	2.73	90.1	2.91
T-LIV	99.0	2.74	98.6	2.63	99.3	2.91
V-LIV	100.0	2.74	99.5	2.63	99.5	2.91
T-FOR	90.0	2.74	87.9	2.67	92.9	2.91
T-TLK	94.4	2.74	93.5	2.63	93.5	2.91
T-P501	85.8	2.74	87.1	2.72	97.1	2.91

is 0.549 for DNSMOS, 0.540 for NISQA, and 0.463 for UTMOS. DNSMOS on T-FOR reaches only 0.360 with 95% CI [0.268, 0.455], and NISQA on its own held-out V-SIM half reaches 0.385 with CI [0.349, 0.418], an equally sub-chance and even tighter interval on the model’s in-sample data. Spearman correlation between $d_M(\mathbf{x})$ and $|\hat{q} - q^*|$ within each subset, a continuous rank-correlation check, agrees qualitatively with the AUROC-based conclusion: correlations are small and inconsistent in sign across predictors and subsets. A single higher-distance-means-higher-error rejection rule is therefore not transferable.

Applying the NISQA-Corpus-fitted Mahalanobis, Euclidean, and nearest-neighbour references, without refitting, to 364 clips from the Interspeech 2025 URGENT Challenge [22] (six languages, 26 enhancement systems) gives domain-detection AUROC up to 0.911 against held-out V-SIM but error-detection AUROC of only 0.415 to 0.541. This dissociation is not specific to NISQA Corpus.

4.4. Conformal Intervals

Table 4 reports the fixed and five-bin Mahalanobis-adaptive intervals. Fixed coverage on held-out V-SIM is 90.2%, 90.1%, and 91.8% for DNSMOS, NISQA, and UTMOS. This confirms the expected in-domain behavior of the disjoint calibration design. Fixed widths span 2.26 to 2.91 MOS, 56–73% of the four-point (1–5) scale, so the coverage guarantee is bought

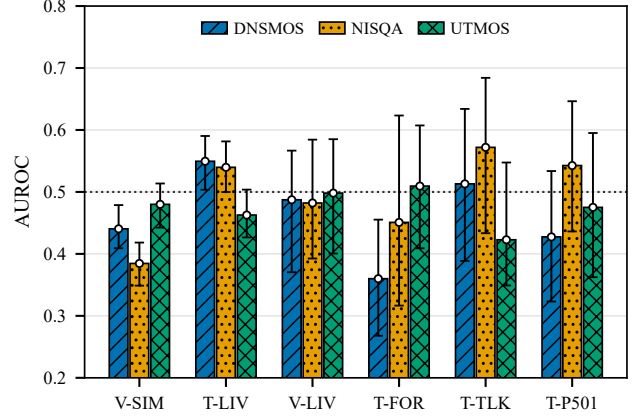


Fig. 1. Within-domain detection of the highest absolute-error quartile using Mahalanobis distance. Color and hatch encode the predictor. Points show AUROC estimates and error bars show group-bootstrap 95% confidence intervals from 2,000 resamples. V-SIM is included because this is error detection rather than domain discrimination.

with intervals that occupy most of the usable range. Coverage under shift ranges from 66.4% to 100.0%, which is consistent with the absence of an arbitrary-shift guarantee.

Adaptation is predictor and domain dependent. For NISQA, coverage changes from 97.1% to 90.4% on T-P501 and from 93.5% to 84.1% on T-TLK. Both changes have bootstrap CIs that exclude zero. k -NN and Euclidean bins avoid this T-TLK undercoverage. The evidence does not support a better wrapper.

5. CONCLUSION

Feature-space distance separates some corpus shifts, but its association with prediction error is weak and can reverse direction. Mahalanobis-stratified intervals therefore improve some predictor–domain pairs and harm others. This dissociation also appears on an independent corpus.

Feature-distance alarms indicate corpus mismatch, not unreliable predictions without predictor- and domain-specific validation. Fixed split conformal intervals provide matched-domain marginal coverage under exchangeability and independent calibration. Arbitrary shift has no coverage guarantee.

Scorer fidelity is a limitation: our UTMOS reimplement and DNSMOS segment-length approximation have unverified equivalence to their reference implementations, and results may differ for other predictors, distances, or domains. Future work should test learned, error-supervised difficulty scores and calibration conditioned on validated domain labels, rather than unsupervised feature distance alone.

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